DINOv2 Classification Heads on DermaMNIST 224×224

Date: 2026-04-20 | **GPU:** NVIDIA A100-SXM4-80GB | **Runtime:** ~22 min

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Setup

Frozen **DINOv2-base** backbone extracts features from 224×224 dermoscopy images. Five classification heads are trained on top of the cached features — no backbone fine-tuning.

- **Dataset:** DermaMNIST — 7,007 train / 1,003 val / 2,005 test, 7 classes
- **CLS heads (MLP, Residual MLP):** use the 768-dim global CLS token
- **Patch heads (CNN Head, CNN Head v2, Attn Pool):** use 256×768 patch tokens reshaped to 768×16×16 (CNN) or kept as sequence (Attn)

Training Curves

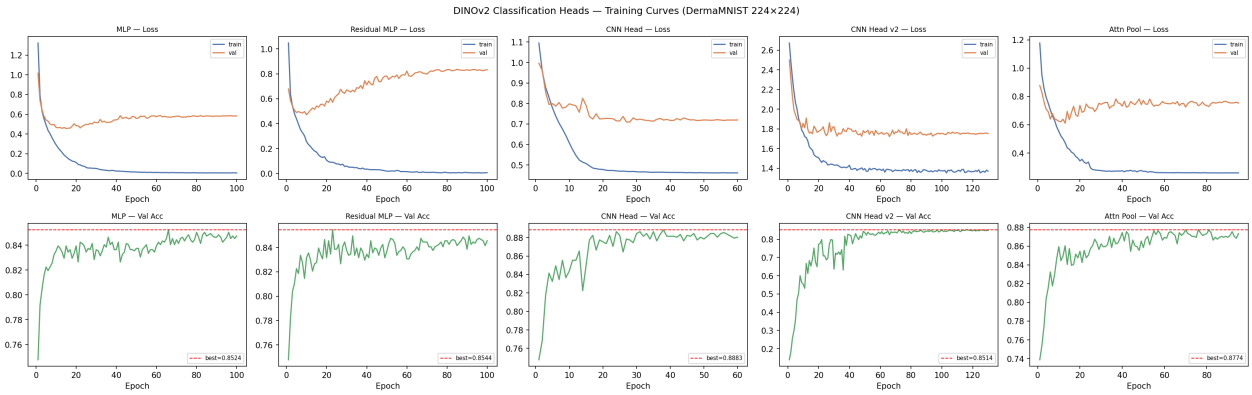


Figure 1: Training Curves

MLP — Clean convergence. Val loss stabilizes early (~epoch 15) while train loss continues to decay, indicating mild overfitting. Val accuracy improves slowly but steadily via cosine schedule, reaching best 0.8524.

Residual MLP — Similar pattern to MLP but val loss diverges more aggressively after epoch 20 (stronger overfitting from extra parameters). Despite this, val accuracy still creeps up due to residual connections, best 0.8544.

CNN Head — Most stable curve of all five. Label smoothing (0.1) keeps train loss above 0.46 throughout, acting as a natural regularizer. Val accuracy rises quickly to ~0.88 by epoch 30 and stays flat — no overfitting. Best 0.8883.

CNN Head v2 — Slowest and noisiest. The 5-epoch warmup and weighted cross-entropy push initial loss to ~2.5. Val accuracy is highly unstable through the first 50 epochs, eventually stabilizing

around 0.85. Early stopped at epoch 130. Best 0.8514 — worse than plain CNN Head despite more complex training.

Attn Pool — Fast initial rise (val acc 0.86 by epoch 10). Plateaus around 0.87–0.88 with noisy fluctuations. Val loss increases throughout while accuracy holds, suggesting the model is overconfident without degrading prediction quality. Early stopped at epoch 95. Best 0.8774.

Overall Performance (Test Set)

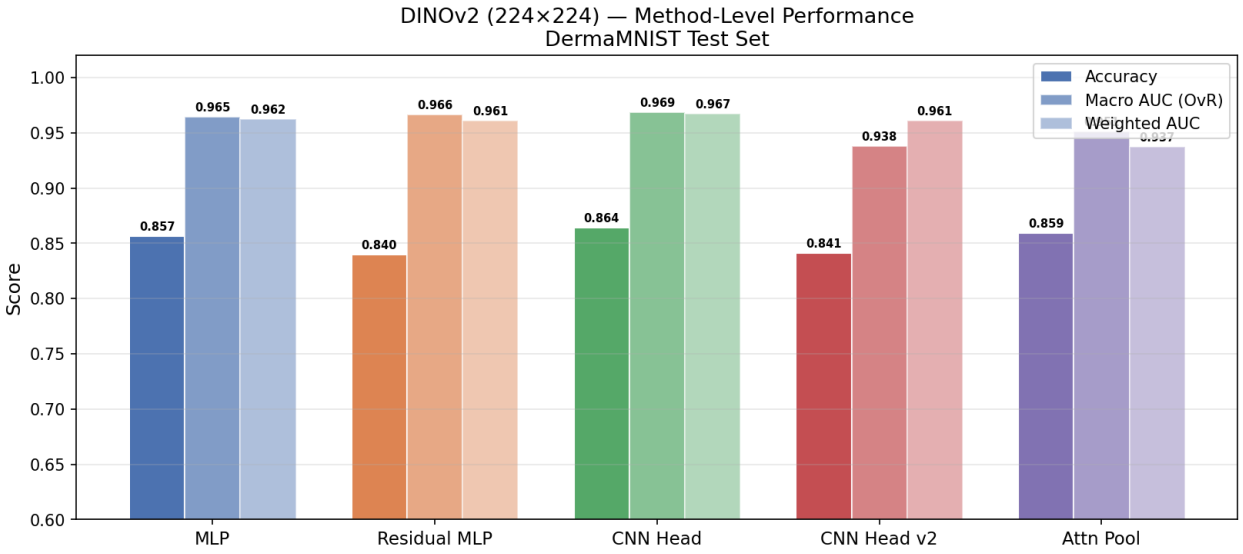


Figure 2: Method-Level Performance

Method	Accuracy	Macro AUC	Weighted AUC	Macro F1
MLP	0.857	0.965	0.962	0.753
Residual MLP	0.840	0.966	0.961	0.712
CNN Head	**0.864**	**0.969**	**0.967**	0.752
CNN Head v2	0.841	0.938	0.961	0.681
Attn Pool	0.859	0.951	0.937	**0.765**

CNN Head leads on accuracy (86.4%), Macro AUC (96.9%), and Weighted AUC (96.7%). Its patch-based convolutional design effectively exploits the spatial structure preserved in DINOv2 patch tokens.

MLP is the strongest CLS-based method — 85.7% accuracy with minimal architecture, confirming the DINOv2 CLS token alone is highly discriminative.

CNN Head v2 underperforms its simpler counterpart despite the class-weighted loss design. The unstable optimization (noisy training curve, early stopping at epoch 130) suggests the weighted loss created conflicting gradient signals that hurt overall generalization.

Attn Pool achieves the best Macro F1 (0.765), reflecting better handling of minority classes via its attention mechanism. However, its AUC scores are noticeably lower (Macro AUC 0.951, Weighted AUC 0.937), especially compared to CNN Head — suggesting less well-calibrated probability outputs.

Residual MLP is the weakest overall. The extra capacity of the ResBlocks does not help with only 7K training samples; it mainly adds overfitting risk.

Per-Class AUC

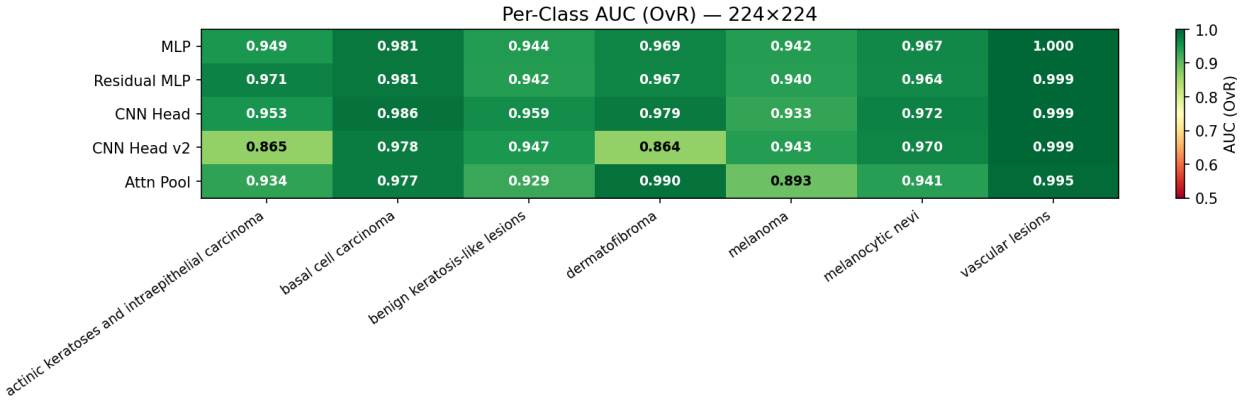
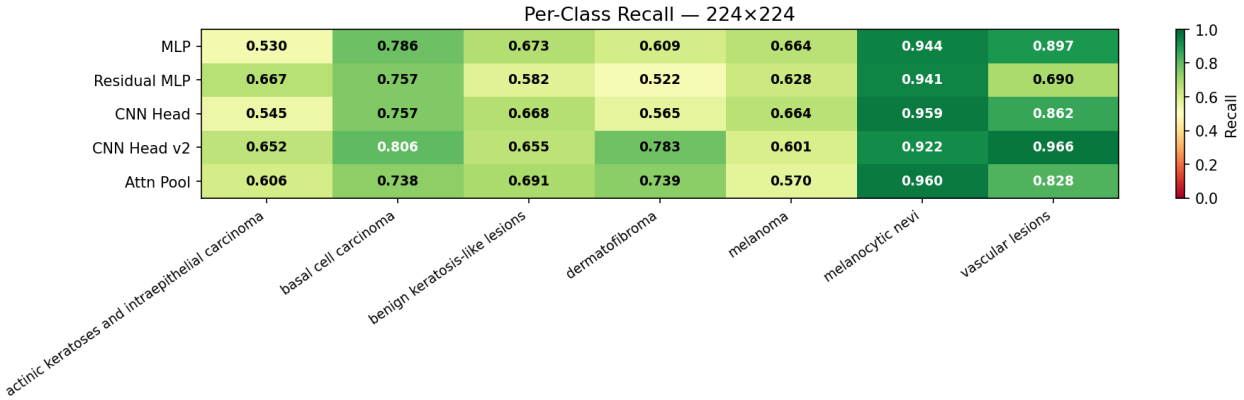


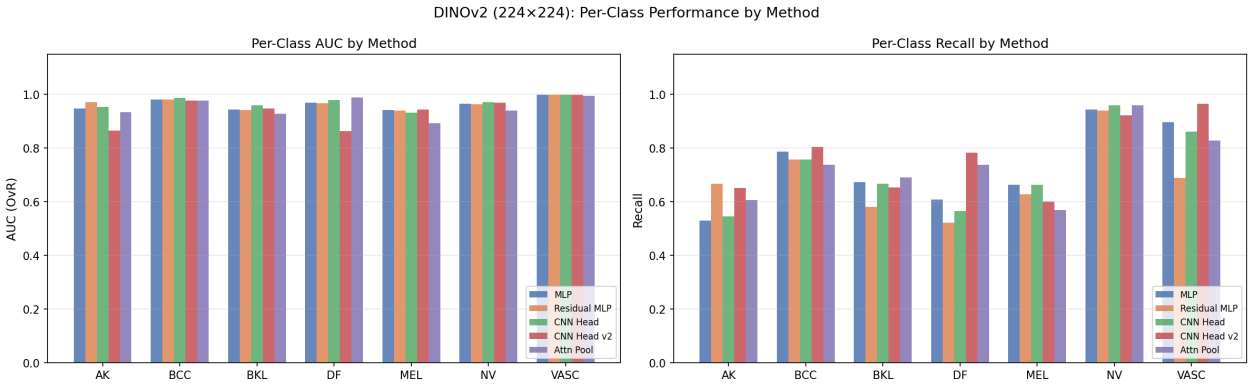
Figure 3: Per-Class AUC Heatmap

All methods achieve very high AUC (>0.93) on most classes, showing DINOv2 features are generally excellent for this task. Notable observations:

- **Vascular lesions (VASC):** near-perfect AUC (0.995–1.000) across all methods — the most visually distinctive class.
- **Basal cell carcinoma (BCC):** consistently high (~0.978–0.986), well-separated in feature space.
- **Melanoma (MEL):** the most clinically critical class, AUC ranges 0.893–0.943. CNN Head v2 achieves a competitive 0.943 here.
- **Actinic keratoses (AK):** CNN Head v2 drops significantly to **0.865**, the worst single score in the entire table — the weighted loss did not help this minority class at the AUC level. MLP and Residual MLP both score >0.949.
- **Dermatofibroma (DF):** CNN Head v2 again drops to 0.864, while Attn Pool scores highest at 0.990 — attention pooling captures local features relevant to this class particularly well.

Per-Class Recall





Recall reveals where each method struggles with actual predictions (not just ranking):

- **Melanocytic nevi (NV):** uniformly high recall (0.922–0.960) — dominant class, easy to predict.
- **Vascular lesions (VASC):** high recall (0.828–0.966); CNN Head v2 achieves the highest at 0.966 — the weighted loss specifically helped this minority class at recall level.
- **Actinic keratoses (AK):** the hardest class across the board. Best recall is only 0.667 (Residual MLP); CNN Head manages only 0.545. All methods are weakest here.
- **Dermatofibroma (DF):** CNN Head v2 achieves the best recall (0.783) and Attn Pool follows (0.739), while CNN Head drops to just 0.565. The spatial patch features benefit attention/weighted approaches for this class.
- **Melanoma (MEL):** recall is moderate (0.570–0.664) across all methods — a concerning finding clinically. No method reliably detects melanoma. CNN Head v2 is worst here (0.601) despite its class-weighting.
- **BKL (Benign keratosis-like lesions):** mid-range recall (0.582–0.691), relatively consistent across methods.

Summary

Criterion	Best Method
Accuracy	CNN Head (86.4%)

Criterion	Best Method
Macro AUC	CNN Head (96.9%)
Weighted AUC	CNN Head (96.7%)
Macro F1	Attn Pool (76.5%)
MEL recall	MLP / CNN Head (tied, 66.4%)
DF recall	CNN Head v2 (78.3%)
Training stability	CNN Head

CNN Head is the best overall method: highest accuracy, highest AUC, and the most stable training. Treating DINOv2 patch tokens as a 16×16 spatial feature map is the most effective inductive bias for this task.

Attn Pool is worth pursuing if minority-class sensitivity (Macro F1) matters more than overall accuracy. Its attention pooling handles DF and AK better in recall terms.

The most important clinical gap across all methods is **low melanoma recall (57–66%)** — none of the frozen-head approaches are reliable enough for MEL detection, suggesting end-to-end fine-tuning of the backbone may be necessary for clinical deployment.